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(II Year, M. Tech. I Sem - Power Electronics and Drives)

Design and Analysis of Radial Flux Permanent Magnet Synchronous Motors Using Ansys Maxwell

Submitted by

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CERTIFICATE

This is to certify that the Summer Internship Report entitled **"Design and Analysis of Radial Flux and Axial Flux Permanent Magnet Synchronous Motors Using Ansys Maxwell"** submitted by **KILLARI DHANUNJAY (Roll No.: 25EEM1R06)** in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Power Electronics & Drives** at the **Department of Electrical Engineering, National Institute of Technology Warangal** is a record of the original work carried out by him under my supervision during the Summer Internship.

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DECLARATION

I, **Killari Dhanunjay**, hereby declare that the project work titled **"Design and Analysis of Radial Flux and Axial Flux Permanent Magnet Synchronous Motors Using Ansys Maxwell"** submitted to the National Institute of Technology Warangal is an original work carried out by me under the guidance of **Dr. Sharmili Das**. This work has not been submitted previously for the award of any degree or diploma at this or any other institution.

Signature

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APPROVAL SHEET

An internship work entitled

Design and Simulation of an H-Bridge Converter and Optimisation of an Isolated DC-DC Converter for Gate-Driver Power supply

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ABSTRACT

This summer internship focused on the design and electromagnetic analysis of a Surface-Mounted Permanent Magnet Synchronous Motor (SPMSM) with a radial flux configuration using Ansys Maxwell 2D. The work involved designing the motor by selecting appropriate electrical and geometrical parameters, including stator and rotor dimensions, permanent magnet arrangement, air-gap length, and winding configuration. A two-dimensional finite element model was developed in Ansys Maxwell 2D to investigate the electromagnetic performance of the machine under rated operating conditions. The simulation study included the analysis of magnetic flux distribution, magnetic flux density, air-gap flux density, back electromotive force (Back EMF), electromagnetic torque, torque ripple, speed characteristics, and phase inductances. The obtained results demonstrated that the designed motor achieved satisfactory electromagnetic performance and met the expected design objectives. This internship provided practical exposure to electric machine design, finite element modelling, electromagnetic simulation, and performance evaluation, while strengthening the understanding of permanent magnet synchronous motor design and finite element analysis techniques using Ansys Maxwell 2D.

Keywords: Surface-Mounted Permanent Magnet Synchronous Motor (SPMSM), Radial Flux PMSM, Ansys Maxwell 2D, Finite Element Analysis (FEA), Back EMF, Electromagnetic Torque.

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List of Abbreviations

PMSM Permanent Magnet Synchronous Motor

SPMSM Surface-Mounted Permanent Magnet Synchronous Motor

AFPM Axial Flux Permanent Magnet Motor

RFPM Radial Flux Permanent Magnet Motor

FEM Finite Element Method

EMF Electromotive Force

SPWM Sine Pulse Width Modulation

SVPWM Space Vector Pulse Width Modulation

Chapter 1

Introduction

1.1 Radial Flux Surface-Mounted Permanent Magnet Synchronous Motor

The Radial Flux Surface-Mounted Permanent Magnet Synchronous Motor (RFSPMSM) is one of the most widely used electrical machines in modern industrial applications due to its high efficiency, compact structure, high power density, and excellent dynamic performance. In this type of machine, permanent magnets are mounted directly on the outer surface of the rotor, while the stator consists of laminated steel cores carrying three-phase distributed windings. The magnetic flux produced by the permanent magnets travels radially across the air gap between the rotor and the stator, giving the machine its name, radial flux machine. The simple rotor construction and the absence of rotor windings reduce losses and improve the overall efficiency of the machine.

When a balanced three-phase alternating current is supplied to the stator windings, a rotating magnetic field is produced. The permanent magnets mounted on the rotor generate a constant magnetic field that interacts with the rotating stator field to produce electromagnetic torque. As the magnetic fields remain synchronized, the rotor rotates at the same speed as the rotating magnetic field without any slip. This synchronous operation provides smooth torque production, excellent speed regulation, and superior dynamic performance. Owing to these advantages, RFSPMSMs are extensively used in electric vehicles, industrial drives, robotics, aerospace systems, renewable energy applications, machine tools, and high-performance servo systems.

The demand for high-efficiency electric machines has increased significantly due to the rapid growth of electric transportation and industrial automation. Compared with conventional induction motors, radial flux permanent magnet synchronous motors offer higher efficiency, improved power factor, better torque density, and lower maintenance requirements. These characteristics make RFSPMSMs an attractive choice for applications requiring compact size, precise speed control, and reliable operation.

1.2 Construction and Operating Principle

A typical radial flux Surface-Mounted Permanent Magnet Synchronous Motor consists of a stator, rotor, permanent magnets, shaft, bearings, and an air gap. The stator is constructed using laminated silicon steel sheets to minimize eddy current losses and contains three-phase distributed windings placed inside uniformly distributed stator slots. The rotor is fabricated from magnetic steel and carries high-energy permanent magnets mounted on its outer surface. These permanent magnets provide the excitation magnetic field without requiring an external DC excitation system.

The operating principle of the RFSPMSM is based on the interaction between the rotating magnetic field produced by the stator windings and the magnetic field established by the permanent magnets on the rotor. When balanced three-phase currents are supplied to the stator, a rotating magnetic field is generated at synchronous speed. The permanent magnets continuously align themselves with this rotating magnetic field, causing the rotor to rotate synchronously without slip. Since there is no rotor copper loss, the efficiency of the machine is considerably improved. The absence of brushes and slip rings further enhances reliability and reduces maintenance requirements.

1.3 Mathematical Model of PMSM

The dynamic performance of a Permanent Magnet Synchronous Motor is generally analysed using the rotating direct-quadrature (d-q) reference frame. The three-phase stator quantities are transformed into two orthogonal components known as the direct-axis (d-axis) and quadrature-axis (q-axis), which rotate synchronously with the rotor magnetic field. This transformation simplifies the mathematical analysis and facilitates the study of machine behaviour under both steady-state and transient operating conditions.

The stator voltage equations in the rotating reference frame are expressed as

$$V_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \tag{1.1}$$

$$V_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \lambda_m \tag{1.2}$$

where V_d and V_q are the direct-axis and quadrature-axis voltages, R_s is the stator resistance, L_d and L_q are the direct-axis and quadrature-axis inductances, i_d and i_q are the corresponding stator currents, ω_e is the electrical angular speed, and λ_m represents the permanent magnet flux linkage.

The electromagnetic torque developed by the PMSM is given by

$$T_e = \frac{3}{2}\frac{P}{2}\left[\lambda_m i_q + (L_d - L_q) i_d i_q\right] \tag{1.3}$$

where P denotes the number of poles.

For a Surface-Mounted Permanent Magnet Synchronous Motor, the rotor is magnetically symmetrical and the direct-axis and quadrature-axis inductances are approximately equal.

$$L_d \approx L_q \tag{1.4}$$

Therefore, the torque equation simplifies to

$$T_e = \frac{3}{2}\frac{P}{2}\lambda_m i_q \tag{1.5}$$

This expression indicates that the electromagnetic torque of a Surface-Mounted PMSM mainly depends on the quadrature-axis current, while the direct-axis current is generally maintained close to zero during normal operation.

1.4 Finite Element Analysis

The electromagnetic performance of electrical machines is strongly influenced by magnetic saturation, leakage flux, fringing effects, and the nonlinear characteristics of magnetic materials. Analytical methods often require simplifying assumptions that may reduce the accuracy of the predicted machine performance. Therefore, the Finite Element Method (FEM) has become the most widely adopted numerical technique for analysing electrical machines. In FEM, the machine geometry is divided into a large number of small finite elements, and Maxwell's electromagnetic field equations are solved numerically over the entire computational domain. This approach enables accurate prediction of magnetic flux distribution, flux density, induced voltage, electromagnetic torque, torque ripple, and inductance while considering nonlinear magnetic material properties. Finite element analysis significantly reduces the need for repeated prototype development and helps optimize machine performance before fabrication.

1.5 Ansys Maxwell 2D

Ansys Maxwell 2D is a finite element analysis software widely used for the design and electromagnetic analysis of electrical machines, transformers, actuators, and other electromagnetic devices. The software provides an integrated environment for creating machine geometry, assigning material properties, defining winding configurations, generating adaptive meshes, specifying boundary conditions, and performing transient electromagnetic simulations. In the present internship, Ansys Maxwell

2D was employed to model the radial flux Surface-Mounted Permanent Magnet Synchronous Motor and evaluate its electromagnetic performance. The simulation results include magnetic flux distribution, magnetic flux density, air-gap flux density, back electromotive force, electromagnetic torque, torque ripple, phase inductances, and other important machine characteristics. These analyses provide valuable insight into the performance of the designed motor and help validate the proposed design.

1.6 Objectives of the Work

The primary objective of this internship is to design and analyse a radial flux Surface-Mounted Permanent Magnet Synchronous Motor using Ansys Maxwell 2D. The work focuses on developing the complete machine geometry based on the specified electrical and geometrical parameters, selecting appropriate magnetic and electrical materials, assigning winding configurations and boundary conditions, and performing transient finite element simulations to evaluate the electromagnetic behaviour of the machine. The study aims to investigate important performance parameters such as magnetic flux distribution, magnetic flux density, back electromotive force, electromagnetic torque, torque ripple, speed characteristics, and phase inductances. The internship also provides practical exposure to finite element modelling and enhances understanding of modern electrical machine design techniques.

1.7 Scope of the Work

The scope of this internship is limited to the electromagnetic design and analysis of a radial flux Surface-Mounted Permanent Magnet Synchronous Motor using Ansys Maxwell 2D. The work includes machine geometry development, material selection, winding configuration, mesh generation, application of boundary conditions, and transient electromagnetic simulation using the finite element method. The performance of the designed motor is evaluated through magnetic flux distribution, magnetic flux density, back electromotive force, electromagnetic torque, torque ripple, and inductance analysis. Thermal analysis, structural analysis, optimization techniques, and experimental validation are beyond the scope of the present work.

1.8 Organization of the Report

This report is organized into four chapters. Chapter 1 introduces the radial flux Surface-Mounted Permanent Magnet Synchronous Motor, its construction, operating principle, mathematical model, finite element analysis, objectives, and scope of the work. Chapter 2 presents the complete design methodology of the machine, including design specifications, machine dimensions, material selection, winding de-

sign, geometry creation, boundary conditions, mesh generation, and solution setup in Ansys Maxwell 2D. Chapter 3 discusses the electromagnetic simulation results, including magnetic flux distribution, magnetic flux density, back electromotive force, electromagnetic torque, torque ripple, speed characteristics, and inductance analysis. Finally, Chapter 4 presents the conclusions drawn from the internship work and outlines possible directions for future research and improvements.

Chapter 2

Machine Design

2.1 Rated Torque and Torque–Speed Requirement

The initial operating requirements of the permanent magnet synchronous motor (PMSM) are specified in terms of the rated output power and rated mechanical speed. The motor is designed for an output power of 5 kW at a rated speed of 2000 rpm. A safety factor of 10% is considered for the preliminary electromagnetic design.

2.1.1 Motor Design Specifications

The initial design specifications are given in Table 2.1.

Table 2.1: Initial design specifications

Parameter	**Symbol**	**Value**
Output power	P_{out}	5 kW
Rated speed	N_{rated}	2000 rpm
DC-link voltage	V_{dc}	600 V
Efficiency	η	95%
Power factor	PF	0.9
Safety factor	SF	10%
Number of poles	$2p$	4
Number of stator slots	Q_s	24
Number of phases	m	3

2.1.2 Rated Mechanical Angular Speed

The mechanical angular speed corresponding to the rated rotational speed is calculated as

$$\omega_m = \frac{2\pi N_{\text{rated}}}{60} \tag{2.1}$$

Substituting the rated speed,

$$\begin{aligned} \omega_m &= \frac{2\pi(2000)}{60} \\ &= 209.44 \text{ rad/s} \end{aligned} \tag{2.2}$$

Therefore,

$$\boxed{\omega_m = 209.44 \text{ rad/s}} \tag{2.3}$$

2.1.3 Rated Operating Torque

The mechanical output power of the motor is related to the electromagnetic torque and mechanical angular speed by

$$P_{\text{out}} = T_{\text{rated}}\omega_m \tag{2.4}$$

Hence, the rated operating torque is

$$T_{\text{rated}} = \frac{P_{\text{out}}}{\omega_m} \tag{2.5}$$

Using the specified output power,

$$\begin{aligned} T_{\text{rated}} &= \frac{5000}{209.44} \\ &= 23.87 \text{ N m} \end{aligned} \tag{2.6}$$

Therefore, the rated operating torque is

$$\boxed{T_{\text{rated}} = 23.87 \text{ N m}} \tag{2.7}$$

2.1.4 Design Torque with Safety Factor

A 10% safety factor is included in the torque calculation to account for design margin and possible operating variations. The design torque is therefore given by

$$T_{\text{design}} = T_{\text{rated}}(1 + SF) \tag{2.8}$$

For a safety factor of 10%,

$$\begin{aligned} T_{\text{design}} &= 23.87(1 + 0.10) \\ &= 26.26 \text{ N m} \end{aligned} \tag{2.9}$$

Thus, the preliminary design torque is

$$\boxed{T_{\text{design}} = 26.26 \text{ N m}} \tag{2.10}$$

2.1.5 Torque–Speed Operating Requirement

For the preliminary motor design, the rated torque is considered constant up to the rated/base speed. Therefore, the design torque in the constant-torque operating region can be expressed as

$$T(N) = T_{\text{design}}, \qquad 0 \le N \le N_{\text{rated}} \tag{2.11}$$

Hence,

$$T(N) = 26.26 \text{ N m}, \qquad 0 \le N \le 2000 \text{ rpm} \tag{2.12}$$

The principal operating points are summarized in Table 2.2.

Table 2.2: Torque–speed design points

Speed (rpm)	**Rated Torque (N m)**	**Design Torque (N m)**
0	23.87	26.26
500	23.87	26.26
1000	23.87	26.26
1500	23.87	26.26
2000	23.87	26.26

2.1.6 Torque–Speed Curve

Figure 2.1 illustrates the preliminary constant-torque operating region with the 10% safety factor.

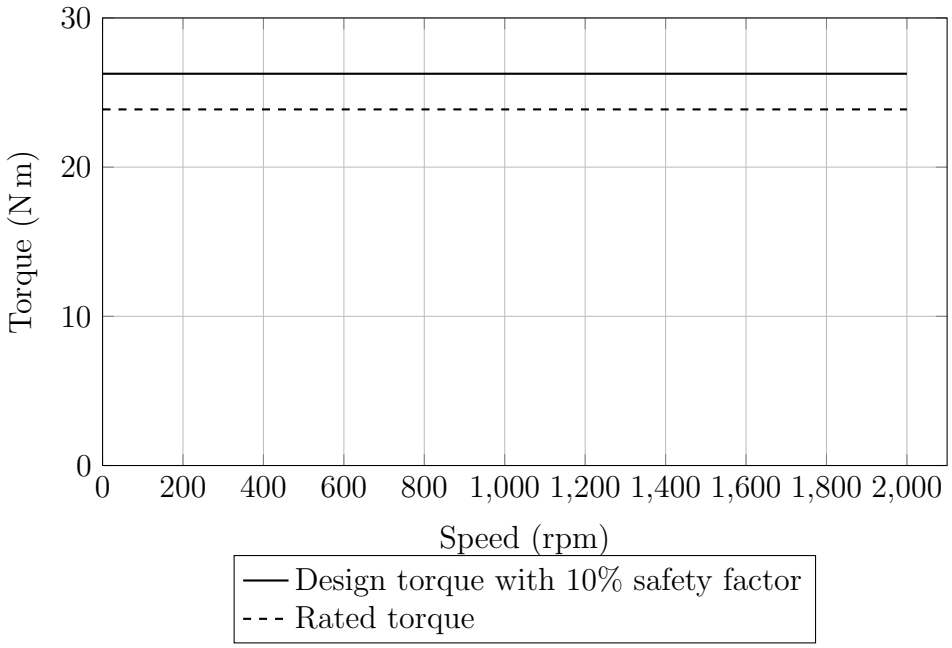


Figure 2.1: Preliminary torque–speed operating requirement

2.1.7 Summary of Torque Calculation

The calculated mechanical quantities used for the subsequent PMSM design calculations are summarized below:

$$\boxed{\begin{aligned} P_{\text{out}} &= 5 \text{ kW} \\ N_{\text{rated}} &= 2000 \text{ rpm} \\ \omega_m &= 209.44 \text{ rad/s} \\ T_{\text{rated}} &= 23.87 \text{ N m} \\ T_{\text{design}} &= 26.26 \text{ N m} \end{aligned}} \tag{2.13}$$

2.2 Selection of Number of Poles and Electrical Frequency

The motor is selected with four poles and a three-phase stator. The number of pole pairs is therefore

$$p = \frac{P}{2} \tag{2.14}$$

For a four-pole machine,

$$p = \frac{4}{2} = 2 \tag{2.15}$$

The relationship between mechanical speed, number of poles, and electrical frequency is given by

$$f_e = \frac{PN}{120} \tag{2.16}$$

where P is the number of poles and N is the mechanical speed in rpm.
For $P = 4$ and $N = 2000$ rpm,

$$\begin{aligned} f_e &= \frac{4(2000)}{120} \\ &= 66.67 \text{ Hz} \end{aligned} \tag{2.17}$$

Therefore,

$$\boxed{f_e = 66.67 \text{ Hz}} \tag{2.18}$$

The corresponding electrical angular frequency is

$$\omega_e = 2\pi f_e \tag{2.19}$$

$$\begin{aligned} \omega_e &= 2\pi(66.67) \\ &= 418.88 \text{ rad/s} \end{aligned} \tag{2.20}$$

Hence,

$$\boxed{\omega_e = 418.88 \text{ rad/s}} \tag{2.21}$$

2.3 Input Active Power

The specified mechanical output power of the motor is

$$P_{\text{out}} = 5 \text{ kW} \tag{2.22}$$

and the desired efficiency is

$$\eta = 95\% = 0.95. \tag{2.23}$$

The input active power is calculated as

$$P_{\text{in}} = \frac{P_{\text{out}}}{\eta} \tag{2.24}$$

Therefore,

$$\begin{aligned} P_{\text{in}} &= \frac{5000}{0.95} \\ &= 5263.16 \text{ W} \end{aligned} \tag{2.25}$$

Hence,

$$\boxed{P_{\text{in}} = 5.263 \text{ kW}} \tag{2.26}$$

The total power loss corresponding to the specified efficiency is

$$P_{\text{loss}} = P_{\text{in}} - P_{\text{out}} \tag{2.27}$$

$$\begin{aligned} P_{\text{loss}} &= 5263.16 - 5000 \\ &= 263.16 \text{ W} \end{aligned} \tag{2.28}$$

Thus,

$$\boxed{P_{\text{loss}} = 263.16 \text{ W}} \tag{2.29}$$

2.4 Apparent Power

The specified power factor of the PMSM drive is

$$PF = 0.9 \tag{2.30}$$

The apparent power is obtained from the input active power and the power factor as

$$S = \frac{P_{\text{in}}}{PF} \tag{2.31}$$

Substituting the calculated input power,

$$\begin{aligned} S &= \frac{5263.16}{0.9} \\ &= 5847.95 \text{ VA} \end{aligned} \tag{2.32}$$

Therefore,

$$\boxed{S = 5.848 \text{ kVA}} \tag{2.33}$$

2.5 Reactive Power

The relationship between the power factor and phase angle is

$$PF = \cos\phi \tag{2.34}$$

Hence,

$$\phi = \cos^{-1}(PF) \tag{2.35}$$

For the specified power factor,

$$\phi = \cos^{-1}(0.9) \tag{2.36}$$

The reactive power is calculated as

$$Q = P_{\text{in}} \tan\phi \tag{2.37}$$

or

$$Q = P_{\text{in}} \tan\left(\cos^{-1} PF\right) \tag{2.38}$$

Substituting the calculated input power and power factor,

$$\begin{aligned} Q &= 5263.16 \tan\left(\cos^{-1}(0.9)\right) \\ &= 2552.8 \text{ VAR} \end{aligned} \tag{2.39}$$

Therefore,

$$\boxed{Q \approx 2.553 \text{ kVAR}} \tag{2.40}$$

2.6 Motor Line-to-Line Voltage Rating

The PMSM is supplied from a two-level voltage source inverter with a DC-link voltage of

$$V_{dc} = 600 \text{ V}. \tag{2.41}$$

The fundamental line-to-line RMS voltage available to the motor depends on the modulation technique.

2.6.1 Line-to-Line Voltage using SPWM

For sinusoidal pulse-width modulation (SPWM), the maximum fundamental line-to-line RMS voltage in the linear modulation region is

$$V_{LL,\text{rms}}^{\text{SPWM}} = \frac{\sqrt{3}}{2\sqrt{2}} V_{dc} \tag{2.42}$$

Substituting the DC-link voltage,

$$\begin{aligned} V_{LL,\text{rms}}^{\text{SPWM}} &= \frac{\sqrt{3}}{2\sqrt{2}}(600) \\ &= 367.42 \text{ V} \end{aligned} \tag{2.43}$$

Therefore,

$$\boxed{V_{LL,\text{rms}}^{\text{SPWM}} = 367.4 \text{ V}} \tag{2.44}$$

2.6.2 Line-to-Line Voltage using SVPWM

For space-vector pulse-width modulation (SVPWM), the maximum fundamental line-to-line RMS voltage in the linear modulation region is

$$V_{LL,\text{rms}}^{\text{SVPWM}} = \frac{V_{dc}}{\sqrt{2}} \tag{2.45}$$

Therefore,

$$\begin{aligned} V_{LL,\text{rms}}^{\text{SVPWM}} &= \frac{600}{\sqrt{2}} \\ &= 424.26 \text{ V} \end{aligned} \tag{2.46}$$

Hence,

$$\boxed{V_{LL,\text{rms}}^{\text{SVPWM}} = 424.3 \text{ V}} \tag{2.47}$$

Table 2.3: Fundamental line-to-line voltage obtained from the 600 V DC-link

Modulation technique	**Maximum fundamental** $V_{LL,\text{rms}}$
SPWM	367.4 V
SVPWM	424.3 V

2.7 Motor Current Rating

The three-phase apparent power is related to the line-to-line RMS voltage and line current by

$$S = \sqrt{3} V_{LL,\text{rms}} I_{L,\text{rms}} \tag{2.48}$$

Therefore, the line current is

$$I_{L,\text{rms}} = \frac{S}{\sqrt{3} V_{LL,\text{rms}}} \tag{2.49}$$

A 10% safety factor is subsequently included in the current rating.

2.7.1 Current Rating for SPWM

Using the maximum fundamental RMS line-to-line voltage obtained with SPWM,

$$V_{LL,\text{rms}}^{\text{SPWM}} = 367.42 \text{ V} \tag{2.50}$$

the corresponding RMS line current is

$$I_{\text{SPWM}} = \frac{5847.95}{\sqrt{3}(367.42)} \tag{2.51}$$

Therefore,

$$\boxed{I_{\text{SPWM}} = 9.19 \text{ A}} \tag{2.52}$$

Including a 10% safety factor,

$$I_{\text{SPWM,design}} = 1.1 I_{\text{SPWM}} \tag{2.53}$$

$$\begin{aligned} I_{\text{SPWM,design}} &= 1.1(9.19) \\ &= 10.11 \text{ A} \end{aligned} \tag{2.54}$$

Hence,

$$\boxed{I_{\text{SPWM,design}} = 10.11 \text{ A}_{\text{rms}}} \tag{2.55}$$

2.7.2 Current Rating for SVPWM

Using the maximum fundamental RMS line-to-line voltage obtained with SVPWM,

$$V_{LL,\text{rms}}^{\text{SVPWM}} = 424.26 \text{ V} \tag{2.56}$$

the corresponding RMS line current is

$$I_{\text{SVPWM}} = \frac{5847.95}{\sqrt{3}(424.26)} \tag{2.57}$$

Therefore,

$$\boxed{I_{\text{SVPWM}} = 7.96 \text{ A}} \tag{2.58}$$

Including a 10% safety factor,

$$I_{\text{SVPWM,design}} = 1.1 I_{\text{SVPWM}} \tag{2.59}$$

$$\begin{aligned} I_{\text{SVPWM,design}} &= 1.1(7.96) \\ &= 8.76 \text{ A} \end{aligned} \tag{2.60}$$

Hence,

$$\boxed{I_{\text{SVPWM,design}} = 8.76 \text{ A}_{\text{rms}}} \tag{2.61}$$

2.8 Stator Winding Configuration

A three-phase double-layer distributed winding is selected for the PMSM. The stator consists of 24 slots and the winding is provided with two parallel paths per phase.

The principal winding parameters are

$$Q_s = 24 \tag{2.62}$$

$$m = 3 \tag{2.63}$$

$$a = 2 \tag{2.64}$$

where Q_s is the number of stator slots, m is the number of phases, and a is the number of parallel paths per phase.

2.8.1 Number of Coils

For a conventional double-layer winding, one coil is assigned to each stator slot. Therefore, the total number of coils is

$$N_{\text{coil}} = Q_s \tag{2.65}$$

Hence,

$$N_{\text{coil}} = 24 \tag{2.66}$$

The number of coils per phase is

$$N_{\text{coil/ph}} = \frac{Q_s}{m} \tag{2.67}$$

Substituting the number of slots and phases,

$$\begin{aligned} N_{\text{coil/ph}} &= \frac{24}{3} \\ &= 8 \end{aligned} \tag{2.68}$$

Therefore,

$$\boxed{N_{\text{coil/ph}} = 8} \tag{2.69}$$

2.8.2 Number of Parallel Paths

Two parallel paths are selected for each phase:

$$\boxed{a = 2} \tag{2.70}$$

The number of series-connected coils in each parallel path is therefore

$$N_{\text{coil,series/path}} = \frac{N_{\text{coil/ph}}}{a} \tag{2.71}$$

Thus,

$$\begin{aligned} N_{\text{coil,series/path}} &= \frac{8}{2} \\ &= 4 \end{aligned} \tag{2.72}$$

Therefore, each phase consists of two parallel branches, with four coils connected in series in each branch:

$$\boxed{\text{(4 series coils)} \parallel \text{(4 series coils)}} \tag{2.73}$$

2.8.3 Phase Voltage

For the selected star-connected three-phase winding, the relationship between line-to-line voltage and phase voltage is

$$V_{ph} = \frac{V_{LL}}{\sqrt{3}} \tag{2.74}$$

For SPWM, the maximum fundamental line-to-line RMS voltage is

$$V_{LL,\text{rms}}^{\text{SPWM}} = 367.4 \text{ V} \tag{2.75}$$

Therefore,

$$\begin{aligned} V_{ph}^{\text{SPWM}} &= \frac{367.4}{\sqrt{3}} \\ &= 212.13 \text{ V} \end{aligned} \tag{2.76}$$

Hence,

$$\boxed{V_{ph}^{\text{SPWM}} = 212.13 \text{ V}} \tag{2.77}$$

For SVPWM,

$$V_{LL,\text{rms}}^{\text{SVPWM}} = 424.3 \text{ V} \tag{2.78}$$

Therefore,

$$\begin{aligned} V_{ph}^{\text{SVPWM}} &= \frac{424.3}{\sqrt{3}} \\ &= 244.95 \text{ V} \end{aligned} \tag{2.79}$$

Thus,

$$\boxed{V_{ph}^{\text{SVPWM}} = 244.95 \text{ V}} \tag{2.80}$$

2.8.4 Voltage per Coil

Since four coils are connected in series in each parallel branch, the phase voltage is distributed among the four series-connected coils. The average voltage per coil is therefore

$$V_{\text{coil}} = \frac{V_{ph}}{N_{\text{coil,series/path}}} \tag{2.81}$$

For SPWM,

$$V_{\text{coil}}^{\text{SPWM}} = \frac{212.13}{4} = 53.03 \text{ V} \tag{2.82}$$

Hence,

$$\boxed{V_{\text{coil}}^{\text{SPWM}} = 53.03 \text{ V}} \tag{2.83}$$

For SVPWM,

$$V_{\text{coil}}^{\text{SVPWM}} = \frac{244.95}{4} = 61.24 \text{ V} \tag{2.84}$$

Therefore,

$$\boxed{V_{\text{coil}}^{\text{SVPWM}} = 61.24 \text{ V}} \tag{2.85}$$

2.8.5 Current per Parallel Path

The phase current is divided equally between the two parallel paths. Therefore,

$$I_{\text{path}} = \frac{I_{ph}}{a} \tag{2.86}$$

For SPWM, the design phase current including the 10% safety factor is

$$I_{\text{SPWM,design}} = 10.11 \text{ A} \tag{2.87}$$

Hence,

$$I_{\text{path}}^{\text{SPWM}} = \frac{10.11}{2} = 5.055 \text{ A} \tag{2.88}$$

Therefore,

$$\boxed{I_{\text{path}}^{\text{SPWM}} = 5.055 \text{ A}} \tag{2.89}$$

For SVPWM, the design phase current is

$$I_{\text{SVPWM,design}} = 8.76 \text{ A} \tag{2.90}$$

Therefore,

$$I_{\text{path}}^{\text{SVPWM}} = \frac{8.76}{2} = 4.38 \text{ A} \tag{2.91}$$

Thus,

$$\boxed{I_{\text{path}}^{\text{SVPWM}} = 4.38 \text{ A}} \tag{2.92}$$

Since the coils in each parallel branch are connected in series, the same branch current flows through each coil of that branch:

$$\boxed{I_{\text{coil}} = I_{\text{path}}} \tag{2.93}$$

2.8.6 Final Winding Arrangement

The final winding arrangement for each phase is

$$\boxed{A: \ (A_1 - A_2 - A_3 - A_4) \parallel (A_5 - A_6 - A_7 - A_8)} \tag{2.94}$$

$$\boxed{B: \ (B_1 - B_2 - B_3 - B_4) \parallel (B_5 - B_6 - B_7 - B_8)} \tag{2.95}$$

$$\boxed{C: \ (C_1 - C_2 - C_3 - C_4) \parallel (C_5 - C_6 - C_7 - C_8)} \tag{2.96}$$

The three phases are connected in a star configuration for the voltage calculations presented above.

2.8.7 Summary of Winding Parameters

Table 2.4: Final stator winding parameters

Parameter	**Value**
Number of phases	3
Number of stator slots	24
Winding type	Double-layer distributed
Total number of coils	24
Coils per phase	8
Parallel paths per phase	2
Series coils per parallel path	4
Connection	Star (Y)
SPWM phase voltage	212.13 V
SVPWM phase voltage	244.95 V
SPWM average coil voltage	53.03 V
SVPWM average coil voltage	61.24 V
SPWM design phase current	10.11 A
SVPWM design phase current	8.76 A
SPWM current per path	5.055 A
SVPWM current per path	4.38 A

2.9 Summary of Electrical Design Parameters

The principal electrical and mechanical design parameters obtained from the initial specifications are summarized in Table 2.5.

Table 2.5: Final stator winding parameters

Parameter	**Symbol**	**Calculated Value**
Output power	P_{out}	5 kW
Rated speed	N	2000 rpm
Rated torque	T_{rated}	23.87 N m
Design torque	T_{design}	26.26 N m
Number of poles	P	4
Pole pairs	p	2
Stator slots	Q_s	24
Number of phases	m	3
Electrical frequency	f_e	66.67 Hz
Input active power	P_{in}	5.263 kW
Power factor	PF	0.9
Apparent power	S	5.848 kVA
Reactive power	Q	2.553 kVAR
DC-link voltage	V_{dc}	600 V
SPWM voltage	$V_{LL,\text{rms}}^{\text{SPWM}}$	367.4 V
SVPWM voltage	$V_{LL,\text{rms}}^{\text{SVPWM}}$	424.3 V
SPWM current	I_{SPWM}	9.19 A
SPWM design current	$I_{\text{SPWM,design}}$	10.11 A
SVPWM current	I_{SVPWM}	7.96 A
SVPWM design current	$I_{\text{SVPWM,design}}$	8.76 A

Chapter 3

Geometry Modeling in Ansys Maxwell

3.1 Creating the Geometry

3.1.1 Main Geometrical Design

The initial main dimensions of the PMSM are determined using the fundamental output equation based on the magnetic and electric loadings.

The rated output power and mechanical speed are

$$P_{\text{out}} = 5 \text{ kW} \tag{3.1}$$

$$N = 2000 \text{ rpm} \tag{3.2}$$

The corresponding mechanical speed in revolutions per second is

$$n = \frac{N}{60} = \frac{2000}{60} = 33.333 \text{ rps.} \tag{3.3}$$

For a four-pole machine, the number of pole pairs is

$$p = \frac{P}{2} = \frac{4}{2} = 2. \tag{3.4}$$

The distinction between the number of poles P and pole pairs p is therefore

$$P = 2p. \tag{3.5}$$

The electrical frequency is consequently

$$f_e = \frac{pN}{60} = \frac{2(2000)}{60} = 66.67 \text{ Hz.} \tag{3.6}$$

Selection of Magnetic and Electric Loading

For a salient-pole synchronous/PMSM machine, the recommended average air-gap magnetic loading is approximately

$$0.85 \leq B_{\text{av}} \leq 1.05 \text{ T.} \tag{3.7}$$

A mid-range value is selected as

$$\boxed{B_{\text{av}} = 0.95 \text{ T}}. \tag{3.8}$$

The specific electric loading is selected within the recommended range

$$35 \leq A \leq 65 \text{ kA/m}. \tag{3.9}$$

Therefore, the selected value is

$$\boxed{A = 50 \text{ kA/m}}. \tag{3.10}$$

Winding Factor

For the 24-slot, 3-phase, 4-pole winding,

$$q = \frac{Q_s}{2pm} = \frac{24}{2(2)(3)} = 2. \tag{3.11}$$

The electrical angle between adjacent stator slots is

$$\alpha_s = \frac{180p}{Q_s} = \frac{180(2)}{24} = 15^\circ. \tag{3.12}$$

For a coil pitch of five slots, the winding is short-pitched by one slot. The pitch factor is therefore

$$k_p = \cos\left(\frac{\alpha_s}{2}\right) = \cos(7.5^\circ) = 0.9914. \tag{3.13}$$

The distribution factor is

$$k_d = \frac{\sin(q\alpha_s/2)}{q\sin(\alpha_s/2)} \tag{3.14}$$

$$k_d = \frac{\sin(2 \times 15^\circ/2)}{2\sin(15^\circ/2)} = 0.9914. \tag{3.15}$$

Hence, the winding factor is

$$k_w = k_p k_d \tag{3.16}$$

$$k_w = (0.9914)(0.9914) \approx \boxed{0.983}. \tag{3.17}$$

Motor Output Equation

The preliminary main dimensions are obtained from the motor output equation

$$P_{\text{out}} = GD^2Ln, \tag{3.18}$$

where D is the air-gap diameter, L is the active stack length, n is the mechanical speed in revolutions per second, and G is the output coefficient.

The output coefficient is given by

$$G = 1.11\pi^2 k_w B_{\text{av}} A. \tag{3.19}$$

Using

$$k_w = 0.983, \qquad B_{\text{av}} = 0.95 \text{ T}, \qquad A = 50 \times 10^3 \text{ A/m},$$

the output coefficient becomes

$$G = 1.11\pi^2(0.983)(0.95)(50 \times 10^3) \tag{3.20}$$

$$\boxed{G = 5.115 \times 10^5 \text{ W/(m}^3 \text{ rps)}} \tag{3.21}$$

or

$$\boxed{G = 511.5 \text{ kW/(m}^3 \text{ rps).}} \tag{3.22}$$

Therefore,

$$D^2L = \frac{P_{\text{out}}}{Gn} \tag{3.23}$$

$$D^2L = \frac{5000}{(5.115 \times 10^5)(33.333)} \tag{3.24}$$

$$\boxed{D^2L = 2.9325 \times 10^{-4} \text{ m}^3.} \tag{3.25}$$

Pole Pitch and Aspect Ratio

The pole pitch is

$$\tau_p = \frac{\pi D}{p}. \tag{3.26}$$

The aspect ratio is defined as

$$a_r = \frac{L}{\tau_p}. \tag{3.27}$$

For the initial analytical design, an aspect ratio of

$$\boxed{a_r = 1.0} \tag{3.28}$$

is selected. Hence,

$$L = \tau_p = \frac{\pi D}{p}. \tag{3.29}$$

For $p = 2$,

$$L = \frac{\pi D}{2}. \tag{3.30}$$

Substituting into the D^2L relation,

$$D^2 \left(\frac{\pi D}{2} \right) = 2.9325 \times 10^{-4} \tag{3.31}$$

and therefore

$$D = \left(\frac{2(2.9325 \times 10^{-4})}{\pi} \right)^{1/3}. \tag{3.32}$$

Thus,

$$\boxed{D = 57.15 \text{ mm}}. \tag{3.33}$$

The corresponding active stack length is

$$L = \frac{\pi(0.05715)}{2} \tag{3.34}$$

$$\boxed{L = 89.78 \text{ mm}}. \tag{3.35}$$

The resulting pole pitch is

$$\boxed{\tau_p = 89.78 \text{ mm}}. \tag{3.36}$$

3.1.2 Stator Slot Dimensions

The stator slot dimensions are selected based on the required winding arrangement, slot area, conductor accommodation, and magnetic loading. The proposed PMSM consists of 24 stator slots, three phases, and four poles. A double-layer distributed winding is considered for the machine.

The principal dimensions of the stator slot are shown in Fig. 3.1.

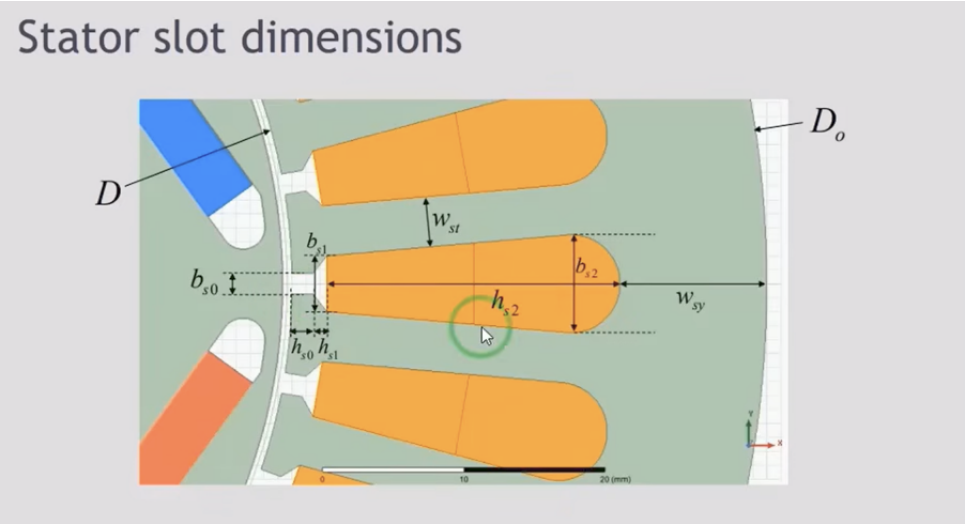


Figure 3.1: Stator slot dimensions of the proposed PMSM

The main slot parameters considered in the geometrical model are shown in Fig. 3.1. These parameters include the slot opening width b_{s0}, slot opening height h_{s0}, slot neck height h_{s1}, slot neck width b_{s1}, slot body width b_{s2}, slot body depth h_{s2}, stator tooth width w_{st}, and stator yoke width w_{sy}.

Stator Slot Pitch

The number of stator slots is

$$Q_s = 24. \tag{3.37}$$

The air-gap diameter obtained from the preliminary machine design is

$$D = 57.15 \text{ mm}. \tag{3.38}$$

The mechanical slot pitch is calculated as

$$\tau_s = \frac{\pi D}{Q_s}. \tag{3.39}$$

Substituting the values,

$$\tau_s = \frac{\pi(57.15)}{24}. \tag{3.40}$$

Therefore,

$$\boxed{\tau_s = 7.481 \text{ mm}} \tag{3.41}$$

The mechanical angular displacement between two adjacent stator slots is

$$\theta_s = \frac{360^\circ}{Q_s}. \tag{3.42}$$

Hence,

$$\theta_s = \frac{360^\circ}{24} = \boxed{15^\circ}. \tag{3.43}$$

Stator Tooth Width

The average air-gap magnetic loading selected for the machine is

$$B_{\text{av}} = 0.95 \text{ T}. \tag{3.44}$$

The active stack length obtained from the preliminary design is

$$L = 89.78 \text{ mm}. \tag{3.45}$$

The number of pole pairs is

$$p = 2. \tag{3.46}$$

The flux per pole is calculated from

$$\Phi_p = B_{\text{av}} \frac{\pi D L}{2p}. \tag{3.47}$$

Substituting the design values,

$$\Phi_p = 0.95 \frac{\pi(0.05715)(0.08978)}{2(2)}. \tag{3.48}$$

Therefore,

$$\boxed{\Phi_p = 3.83 \text{ mWb}} \tag{3.49}$$

The number of stator slots per pole is

$$N_{\text{sp}} = \frac{Q_s}{2p} = \frac{24}{4} = 6. \tag{3.50}$$

The preliminary stator tooth flux density is selected as

$$B_{st} = 1.6 \text{ T}. \tag{3.51}$$

The tooth width is estimated from

$$w_{st} = \frac{\Phi_p / N_{\text{sp}}}{B_{st} L}. \tag{3.52}$$

Substituting the values,

$$w_{st} = \frac{3.83 \times 10^{-3}/6}{1.6(0.08978)}. \tag{3.53}$$

Therefore,

$$\boxed{w_{st} \approx 4.44 \text{ mm}} \tag{3.54}$$

Slot Opening Dimensions

A relatively small slot opening is selected to reduce the slot leakage and maintain an adequate air-gap permeance distribution. The preliminary slot opening width is selected as

$$\boxed{b_{s0} = 1.0 \text{ mm}} \tag{3.55}$$

and the slot opening height is selected as

$$\boxed{h_{s0} = 1.0 \text{ mm.}} \tag{3.56}$$

The slot neck height is selected as

$$\boxed{h_{s1} = 1.5 \text{ mm.}} \tag{3.57}$$

Therefore, the total height of the slot opening and neck is

$$h_{s0} + h_{s1} = 1.0 + 1.5 = \boxed{2.5 \text{ mm}}. \tag{3.58}$$

Slot Neck Width

The slot neck width is calculated from the slot geometry as

$$b_{s1} = 2\left[\tan\left(\frac{\pi}{Q_s}\right)\left(\frac{D}{2} + h_{s0} + h_{s1}\right) - \frac{w_{st}/2}{\cos(\pi/Q_s)}\right]. \tag{3.59}$$

Using

$$D = 57.15 \text{ mm}, \qquad Q_s = 24, \tag{3.60}$$

$$h_{s0} = 1.0 \text{ mm}, \qquad h_{s1} = 1.5 \text{ mm}, \tag{3.61}$$

and

$$w_{st} = 4.44 \text{ mm}, \tag{3.62}$$

the slot neck width becomes

$$b_{s1} = 2\left[\tan\left(\frac{\pi}{24}\right)(28.575 + 1.0 + 1.5) - \frac{4.44/2}{\cos(\pi/24)}\right]. \tag{3.63}$$

Thus,

$$\boxed{b_{s1} \approx 3.70 \text{ mm}} \tag{3.64}$$

Stator Yoke Thickness

The stator yoke thickness is estimated from the required yoke flux density. A preliminary stator yoke flux density of

$$B_{sy} = 1.6 \text{ T} \tag{3.65}$$

is selected.

The required yoke thickness is estimated using

$$h_{sy} = \frac{\Phi_p}{2B_{sy}L}. \tag{3.66}$$

Substituting the values,

$$h_{sy} = \frac{3.83 \times 10^{-3}}{2(1.6)(0.08978)}. \tag{3.67}$$

Therefore,

$$\boxed{h_{sy} \approx 13.33 \text{ mm}} \tag{3.68}$$

Slot Body Dimensions

The slot body dimensions are selected to provide sufficient space for the required stator conductors while maintaining an adequate stator tooth width and yoke thickness.

A preliminary slot body depth is selected as

$$\boxed{h_{s2} = 14.0 \text{ mm}} \tag{3.69}$$

and the preliminary slot body width is selected as

$$\boxed{b_{s2} = 4.5 \text{ mm}.} \tag{3.70}$$

These values are preliminary geometrical dimensions and can be further optimized after determining the required number of conductors, current density, copper

area, and slot fill factor.

Summary of Stator Slot Dimensions

The preliminary stator slot dimensions used for the Ansys Maxwell geometrical model are summarized in Table 3.1.

Table 3.1: Preliminary stator slot dimensions

Parameter	**Symbol**	**Value**
Air-gap diameter	D	57.15 mm
Active stack length	L	89.78 mm
Number of stator slots	Q_s	24
Number of pole pairs	p	2
Slot pitch	τ_s	7.481 mm
Slot mechanical angle	θ_s	15°
Average air-gap flux density	B_{av}	0.95 T
Stator tooth flux density	B_{st}	1.6 T
Stator yoke flux density	B_{sy}	1.6 T
Flux per pole	Φ_p	3.83 mWb
Stator tooth width	w_{st}	4.44 mm
Slot opening width	b_{s0}	1.00 mm
Slot opening height	h_{s0}	1.00 mm
Slot neck height	h_{s1}	1.50 mm
Slot neck width	b_{s1}	3.70 mm
Slot body width	b_{s2}	4.50 mm
Slot body depth	h_{s2}	14.0 mm
Stator yoke thickness	h_{sy}	13.33 mm

The above dimensions provide the initial stator slot geometry for the PMSM model. The final slot dimensions should be verified using the electromagnetic and thermal requirements of the machine. In particular, the slot body dimensions should be checked against the required conductor cross-sectional area and the selected slot fill factor.

3.1.3 Determination of Number of Turns per Coil

The number of turns per coil is determined from the RMS back-EMF equation of the PMSM. The rated supply voltage is considered as

$$V_{\text{ph}} = 230 \text{ V}_{\text{rms}}. \tag{3.71}$$

The required ratio between the back EMF and the applied voltage is assumed to be

$$\frac{E_{\text{ph}}}{V_{\text{ph}}} = 0.91. \tag{3.72}$$

Therefore, the required RMS back EMF is

$$E_{\text{ph}} = 0.91V_{\text{ph}} = 0.91(230) \tag{3.73}$$

$$\boxed{E_{\text{ph}} = 209.3 \text{ V}_{\text{rms}}} \tag{3.74}$$

For a sinusoidal PMSM, the RMS back-EMF equation is

$$E_{\text{ph}} = 4.44 f_e \Phi_p N_{\text{ph}} k_w \tag{3.75}$$

where f_e is the electrical frequency, Φ_p is the flux per pole, N_{ph} is the number of series turns per phase, and k_w is the winding factor.

For the proposed machine,

$$f_e = 66.67 \text{ Hz}, \tag{3.76}$$

$$\Phi_p = 3.83 \times 10^{-3} \text{ Wb}, \tag{3.77}$$

and

$$k_w = 0.983. \tag{3.78}$$

Hence, the required number of series turns per phase is

$$N_{\text{ph}} = \frac{E_{\text{ph}}}{4.44 f_e \Phi_p k_w}. \tag{3.79}$$

Substituting the above values,

$$N_{\text{ph}} = \frac{209.3}{4.44(66.67)(3.83 \times 10^{-3})(0.983)} \tag{3.80}$$

$$\boxed{N_{\text{ph}} \approx 188 \text{ turns/phase}} \tag{3.81}$$

The proposed machine has 24 stator slots and a double-layer winding. Therefore, the total number of coils is 24, giving

$$N_{c,\text{ph}} = \frac{24}{3} = 8 \text{ coils/phase}. \tag{3.82}$$

The theoretical number of turns per coil is consequently

$$N_{\text{coil}} = \frac{N_{\text{ph}}}{N_{c,\text{ph}}} = \frac{188}{8} \approx 23.5. \tag{3.83}$$

Since the number of turns must be an integer, the number of turns per coil is selected as

$$\boxed{N_{\text{coil}} = 24 \text{ turns/coil}} \tag{3.84}$$

Thus, the actual number of series turns per phase becomes

$$N_{\text{ph,actual}} = 24 \times 8 = \boxed{192 \text{ turns/phase}}. \tag{3.85}$$

The resulting back EMF corresponding to 24 turns per coil is

$$E_{\text{ph,actual}} = 4.44(66.67)(3.83 \times 10^{-3})(192)(0.983) \tag{3.86}$$

$$\boxed{E_{\text{ph,actual}} \approx 214 \text{ V}_{\text{rms}}} \tag{3.87}$$

Thus, the initial winding design for the PMSM uses

$$\boxed{24 \text{ turns/coil}} \tag{3.88}$$

with 8 coils per phase and 192 series turns per phase.

Conductors per Slot

Since the machine uses a double-layer winding, each stator slot contains two coil sides. For the selected winding design,

$$N_{\text{coil}} = 24 \text{ turns/coil}. \tag{3.89}$$

Each turn consists of two active conductor sides. Therefore, the number of conductors in one coil side is

$$Z_{\text{side}} = 2N_{\text{coil}} = 2(24) = 48. \tag{3.90}$$

Since two coil sides are accommodated in each slot, the total number of conductors per slot is

$$Z_{\text{slot}} = 2Z_{\text{side}} = 2(48) = \boxed{96 \text{ conductors/slot}}. \tag{3.91}$$

Thus, the selected winding contains

$$\boxed{Z_{\text{slot}} = 96 \text{ conductors/slot}}. \tag{3.92}$$

3.1.4 Slot Area, Conductor Area and Slot Fill Factor

According to the winding design, the PMSM has 24 stator slots, 24 coils, and a double-layer distributed winding. The number of coils per phase is

$$N_{c,\text{ph}} = \frac{24}{3} = 8 \text{ coils/phase.} \tag{3.93}$$

Two parallel paths are used for each phase. From the SPWM design, the phase current is

$$I_{\text{ph}} = 10.11 \text{ A.} \tag{3.94}$$

Therefore, the current carried by each parallel path is

$$I_{\text{path}} = \frac{I_{\text{ph}}}{a} = \frac{10.11}{2} = \boxed{5.055 \text{ A}}. \tag{3.95}$$

A copper current density of

$$J = 5 \text{ A/mm}^2 \tag{3.96}$$

is selected. Hence, the required cross-sectional area of one conductor is

$$A_{\text{cond}} = \frac{I_{\text{path}}}{J} = \frac{5.055}{5} = \boxed{1.011 \text{ mm}^2}. \tag{3.97}$$

The selected number of turns per coil is

$$N_{\text{coil}} = 24 \text{ turns/coil.} \tag{3.98}$$

Since a double-layer winding is used, each slot contains two coil sides. Therefore, the number of conductor cross-sections per slot is

$$Z_{\text{slot}} = 2N_{\text{coil}} = 2(24) = \boxed{48}. \tag{3.99}$$

The total copper area required in one stator slot is

$$A_{\text{cu,slot}} = Z_{\text{slot}} A_{\text{cond}} \tag{3.100}$$

$$A_{\text{cu,slot}} = 48(1.011) = \boxed{48.528 \text{ mm}^2}. \tag{3.101}$$

The gross slot area obtained from the selected stator slot geometry is

$$\boxed{A_{\text{slot,gross}} = 69.55 \text{ mm}^2}. \tag{3.102}$$

The slot fill factor is defined as

$$k_{\text{fill}} = \frac{A_{\text{cu,slot}}}{A_{\text{slot,gross}}}. \tag{3.103}$$

Therefore,

$$k_{\text{fill}} = \frac{48.528}{69.55} = 0.6979. \tag{3.104}$$

Hence, the slot fill factor is

$$\boxed{k_{\text{fill}} = 69.79\%}. \tag{3.105}$$

The resulting winding and slot parameters are

$$\boxed{\begin{aligned} I_{\text{ph}} &= 10.11 \text{ A}, & I_{\text{path}} &= 5.055 \text{ A}, \\ a &= 2, & J &= 5 \text{ A/mm}^2, \\ A_{\text{cond}} &= 1.011 \text{ mm}^2, & N_{\text{coil}} &= 24 \text{ turns/coil}, \\ Z_{\text{slot}} &= 48, & A_{\text{cu,slot}} &= 48.528 \text{ mm}^2, \\ A_{\text{slot,gross}} &= 69.55 \text{ mm}^2, & k_{\text{fill}} &= 69.79\%. \end{aligned}} \tag{3.106}$$

3.2 Material Assignment

The appropriate materials were assigned to the different components of the radial flux Surface-Mounted Permanent Magnet Synchronous Motor (SPMSM) model in Ansys Maxwell. The stator core was assigned M250-35A electrical steel, which is suitable for laminated magnetic cores. NdFeB permanent magnet material was assigned to the rotor magnets to provide the required excitation field. Copper was assigned to the stator windings because of its high electrical conductivity. The remaining non-magnetic regions were assigned suitable air/vacuum material.

The materials used in the electromagnetic model are summarized in Table 3.2.

Table 3.2: Material Assignment

Component	**Assigned Material**
Stator Core	Cogent Power M250-35A
Rotor Permanent Magnets	Arnold Magnetics N42EH
Stator Windings	Copper
Air Region	Vacuum

The material properties were defined in the Ansys Maxwell material library according to the requirements of the electromagnetic analysis. The electrical steel was specified with its magnetic B-H characteristics, while the permanent magnet was defined using its magnetic properties. Copper was assigned its electrical conductivity for the winding region..

3.3 Boundary Conditions

Boundary conditions were applied to define the electromagnetic behavior at the outer boundary of the computational region. In the present Ansys Maxwell 2D transient model, a Vector Potential boundary condition was assigned to the outer boundary of the solution region. This boundary condition limits the magnetic field at the external boundary and allows the electromagnetic field to be solved within the motor region. The boundary was selected sufficiently far from the active machine geometry to minimize its influence on the calculated electromagnetic quantities.

3.4 Excitations

The stator windings were excited using a balanced three-phase sinusoidal voltage supply. Separate winding excitations were assigned to Phase A, Phase B, and Phase C in Ansys Maxwell. The voltage applied to Phase A was defined as

$$V_A(t) = \sqrt{2}\, V_{\text{ph,rms}} \sin(2\pi f t) \tag{3.107}$$

The remaining phase voltages were displaced by 120° electrically and are given by

$$V_B(t) = \sqrt{2}\, V_{\text{ph,rms}} \sin\left(2\pi f t - \frac{2\pi}{3}\right), \tag{3.108}$$

$$V_C(t) = \sqrt{2}\, V_{\text{ph,rms}} \sin\left(2\pi f t + \frac{2\pi}{3}\right). \tag{3.109}$$

In the Maxwell model, the winding excitation was specified as a voltage excitation with an initial current of zero. The winding resistance was set to 1.5 Ω, while the number of parallel branches was set to one. The same excitation approach was applied to the three phases to obtain a balanced rotating magnetic field in the stator.

The three-phase excitation produces the rotating magnetic field required for synchronous operation of the surface-mounted permanent magnet rotor. The resulting interaction between the stator magnetic field and the permanent magnet field produces the electromagnetic torque of the motor.

Chapter 4

Results and Discussion

4.1 Introduction

The designed radial flux Surface-Mounted Permanent Magnet Synchronous Motor (SPMSM) was analyzed using Ansys Maxwell 2D under transient operating conditions. The electromagnetic simulation was performed to evaluate the magnetic field distribution, stator currents, phase voltages, rotor position, d–q axis currents, electromagnetic torque, and rotor speed. The obtained simulation results are presented and discussed in this chapter. The results are used to verify the electromagnetic behavior of the designed machine and to assess its suitability for the specified operating conditions.

4.2 Magnetic Field Analysis

The magnetic field distribution obtained from the transient electromagnetic analysis is shown in Fig. 4.1. The magnetic flux is established between the surface-mounted permanent magnets on the rotor and the stator teeth and yoke. The field distribution indicates that the magnetic flux crosses the air gap in the radial direction and links with the stator windings.

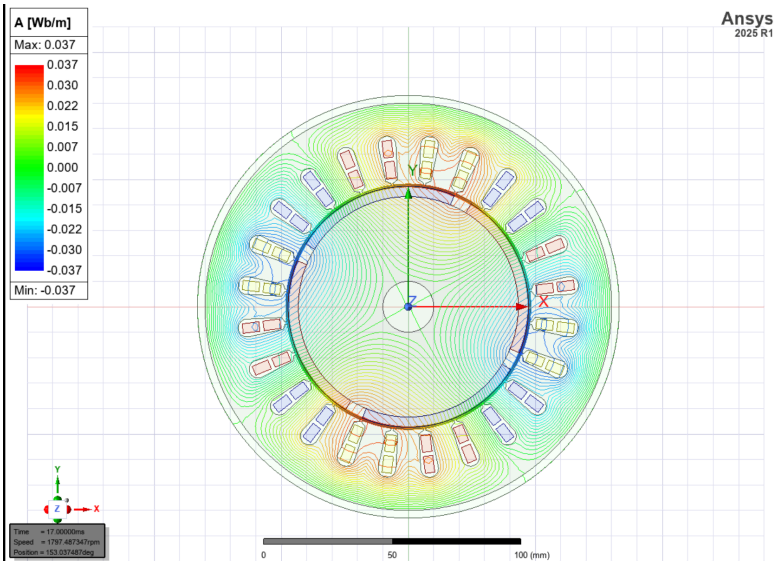


Figure 4.1: Magnetic flux distribution of the radial flux SPMSM

The resultant magnetic field distribution obtained from Ansys Maxwell is shown in Fig. 4.2. The field pattern confirms the interaction between the permanent magnet

field and the stator magnetic field. The magnetic field is distributed around the air gap and stator teeth, corresponding to the four-pole rotor configuration.

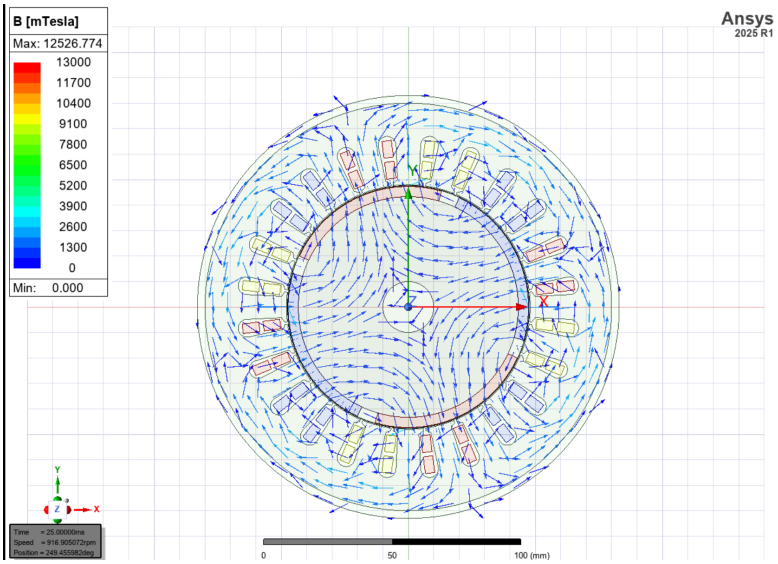


Figure 4.2: Resultant magnetic field distribution

4.3 Back Electromotive Force (Back EMF)

The three-phase stator voltages obtained from the transient simulation are shown in Fig. 4.3. The phase voltages are approximately sinusoidal and are displaced from each other by approximately 120° electrical. This balanced voltage waveform is obtained from the interaction between the rotating permanent magnet field and the stator windings.

The induced phase voltage can be expressed as

$$e(t) = N\frac{d\Phi}{dt} \tag{4.1}$$

where $e(t)$ is the induced back EMF, N is the effective number of turns, and Φ is the flux linking the winding.

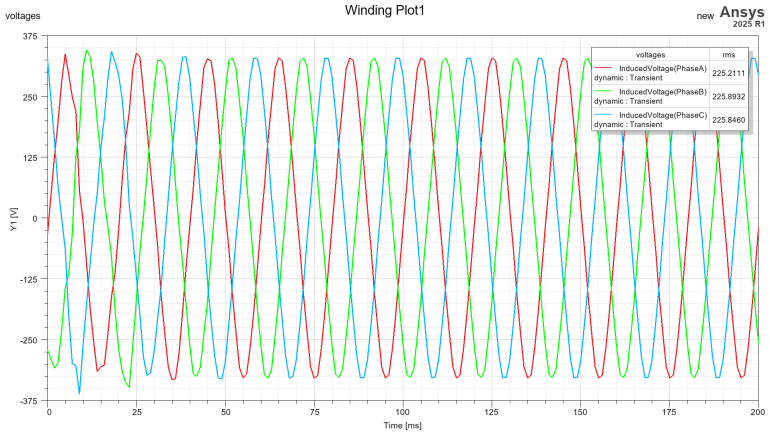


Figure 4.3: Three-phase stator voltages

The simulated voltage waveform provides an indication of the quality of the magnetic flux distribution and winding arrangement. A nearly sinusoidal back EMF is desirable because it reduces unwanted harmonic components and supports smooth torque production.

4.4 Stator Current Analysis

The simulated three-phase stator currents are shown in Fig. 4.4. The currents are balanced and approximately sinusoidal after the initial transient period. The initial oscillations are associated with the starting transient of the machine. As the motor reaches steady-state operation, the current waveforms become more stable.

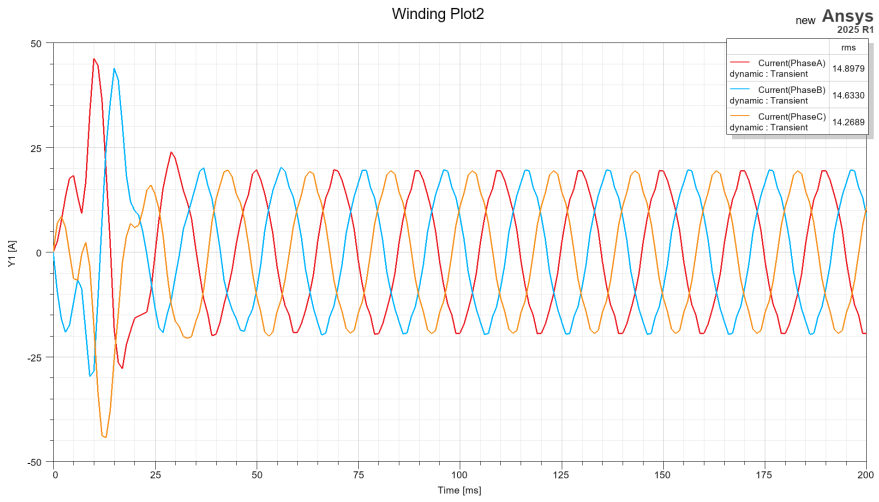


Figure 4.4: Three-phase stator currents

The balanced nature of the phase currents indicates that the three-phase winding excitation is correctly established in the Maxwell model. The current waveform is also important for evaluating the electromagnetic torque and copper loss of the machine.

4.5 d–q Axis Current Analysis

The d–q axis current components obtained from the simulation are shown in Fig. 4.5. The d-axis current represents the flux-producing component, while the q-axis current is mainly responsible for electromagnetic torque production.

For a surface-mounted PMSM, the electromagnetic torque can be expressed as

$$T_e = \frac{3}{2} p \left[\psi_f i_q + (L_d - L_q) i_d i_q \right] \tag{4.2}$$

where p is the number of pole pairs, ψ_f is the permanent magnet flux linkage, L_d and L_q are the d- and q-axis inductances, and i_d and i_q are the corresponding current components.

For a surface-mounted PMSM, L_d and L_q are generally close in value. Therefore, the reluctance torque component is relatively small, and the electromagnetic torque is mainly produced by the interaction between the permanent magnet flux and the q-axis current.

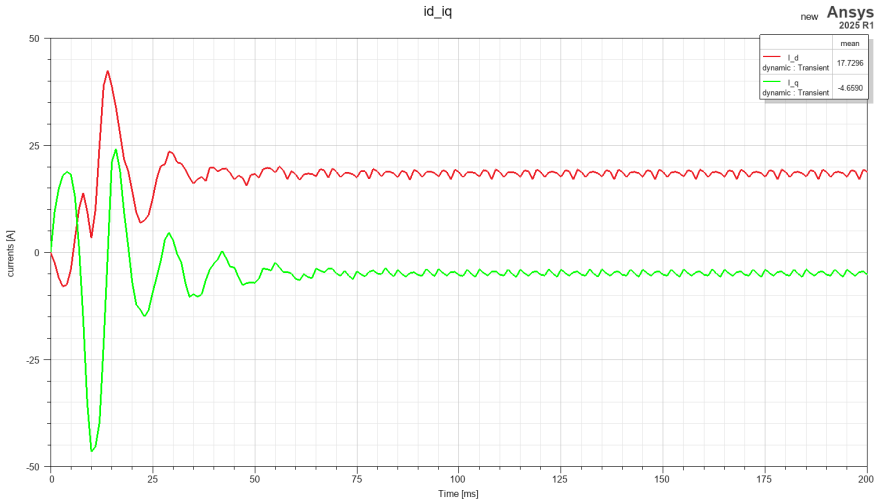


Figure 4.5: d-axis and q-axis current components

4.6 Electromagnetic Torque Analysis

The electromagnetic torque obtained from the transient simulation is shown in Fig. 4.6. The torque response contains an initial transient component followed by a comparatively steady torque region. The initial oscillations are produced during the starting transient as the rotor synchronizes with the rotating magnetic field.

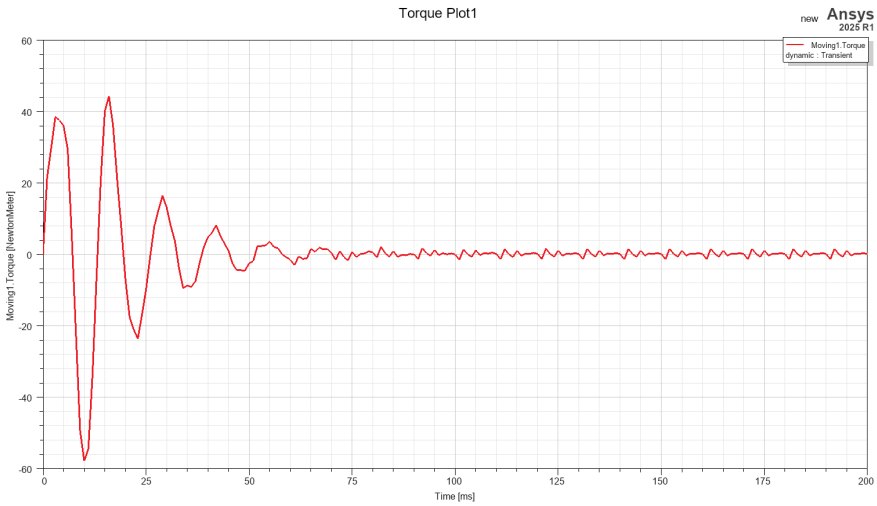


Figure 4.6: Electromagnetic torque response

After the transient period, the electromagnetic torque approaches its steady-state operating value. Small variations in the steady-state torque are associated with

torque ripple resulting from slotting effects, winding distribution, and the interaction between the rotor magnets and stator magnetic field.

The average electromagnetic torque can be calculated from

$$T_{\text{avg}} = \frac{1}{T} \int_0^T T_e(t)\, dt \tag{4.3}$$

where $T_e(t)$ represents the instantaneous electromagnetic torque and T is the time interval considered for averaging.

4.7 Rotor Speed and Position

The simulated rotor speed response is shown in Fig. 4.7. The speed exhibits an initial transient response followed by a steady operating region. The initial variation is caused by the electromagnetic acceleration of the rotor from the initial condition.

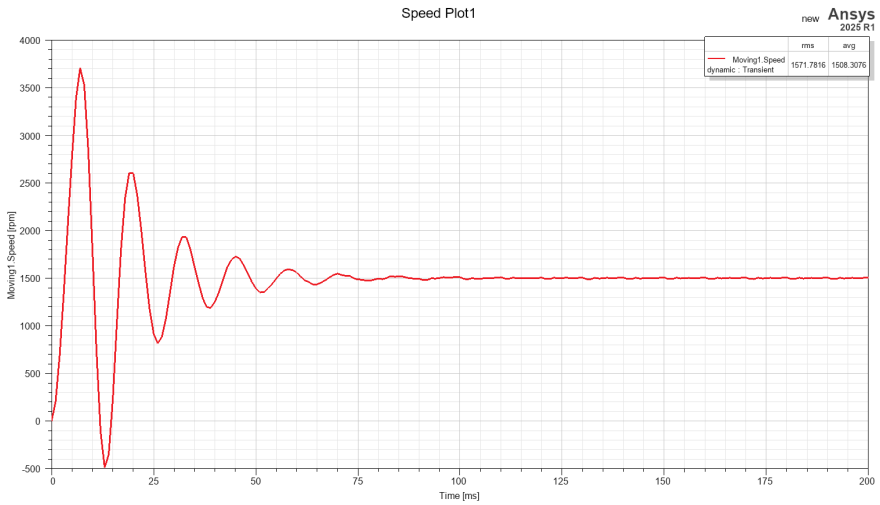


Figure 4.7: Rotor speed response

Once the electromagnetic torque and mechanical dynamics reach equilibrium, the rotor speed approaches the commanded operating speed. The steady-state speed response indicates the synchronous operation of the PMSM.

The mechanical angular speed is related to the rotor speed by

$$\omega_m = \frac{2\pi N}{60} \tag{4.4}$$

where N is the rotor speed in rpm.

The simulated rotor position is shown in Fig. 4.8. The position increases continuously with time, indicating continuous rotor rotation. The periodic variation of the position is consistent with the rotational operation of the four-pole machine.

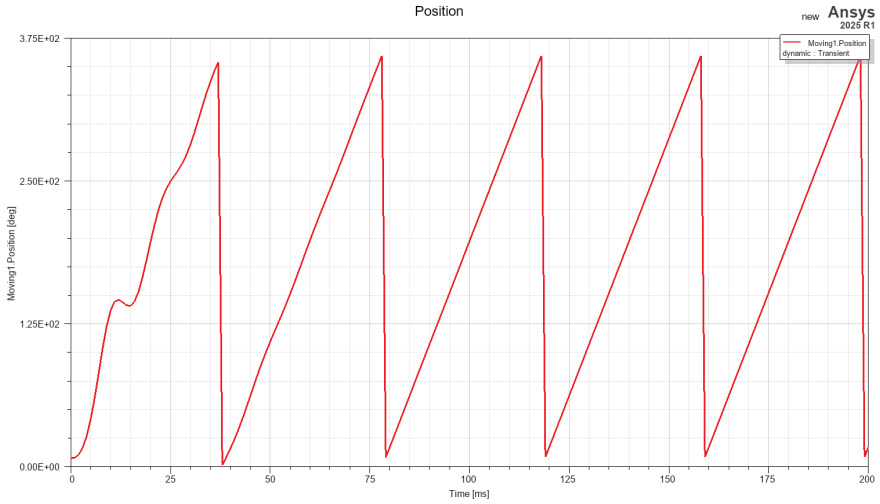


Figure 4.8: Rotor position response

4.8 Performance Analysis

The overall simulation results confirm the expected electromagnetic behavior of the designed radial flux SPMSM. The magnetic field distribution shows proper interaction between the permanent magnets and stator windings. The three-phase voltage and current waveforms demonstrate balanced operation, while the d–q current response provides information about the flux- and torque-producing current components.

The electromagnetic torque reaches a steady operating region after the initial transient, and the rotor speed approaches synchronous operation. These results indicate that the designed machine and the corresponding Ansys Maxwell model are functioning according to the intended operating principle.

Overall, the transient electromagnetic analysis demonstrates that the designed radial flux SPMSM produces the required magnetic field, balanced three-phase electrical quantities, electromagnetic torque, and synchronous rotor motion. The obtained results provide the basis for further evaluation of torque ripple, losses, efficiency, and other performance characterstics.

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